Magnetism in Matter I think I can safely say that nobody understands quantum mechanics.*Richard Feynm*
[width=0.5] /PhyAI/chapter6/6_1*.png* A loop inside magnetic field $\vec{B}$ A rectangular loop has size $a \times b$, under the net torqu

I define $\vec{n}$ as the normal vector of the loop, and an angle $\theta = (\vec{n}, \vec{B})$, because both torques vectors are in the same direction.
[width=0.5] /PhyAI/chapter6/6_2*.png* Both torque vectors in pointing out from the plane of paper The scalar τ_{net} value is:

I define a new vector $\vec{\mu}$ quantity, called **magnetic dipole moment**, same direction to normal vector $\vec{n}$, and has magnitude

Rewriting the net torque equation yields:

[Magnetic Moment Definition] The **magnetic moment** (or magnetic dipole moment) is a vector quantity that determines t

Each angle θ (position), it is characterized by a **potential energy value**. I randomly choose 2 angles θ_i and θ_f:

I assign the potential energy value to angle θ:

From here we have a very important result: a magnetic dipole has its lowest energy when its vector $\vec{\mu}$ is lined up with magn
[width=0.666666] /PhyAI/chapter6/6_3*.png* Highest and lowest potential energy value To flip an arrow (magnetic moment v
[width=1] /PhyAI/chapter6/6_4*.png* A counterclockwise current inside a diverging magnetic field
[H]
[width=1] /PhyAI/chapter6/6_5*.png* A clockwise current inside a diverging magnetic field [H]
[width=0.95] /PhyAI/chapter6/6_6*.png* Two loops system tends to be **repelled away** from the north pole Inside Bohr's ato
[width=0.75] /PhyAI/chapter6/6_7*.png* The macroscopic view of paramagnetic material An external magnetic field $\vec{B}$ imme

In 1985, Pierre Curie discovered experimentally the relationship:

This proportional relationship seems like very obvious; the higher the temperature, the harder the alignment of magnetic m
[width=0.5] /PhyAI/chapter6/6_8*.png* Two-state paramagnet If I denote N as "the number of", the multiplicity and entrop

The internal energy is determined by:

I neglect the constant B in the magnetization formula, and redefine the M quantity as:

Before we look for an analytical solution, let's consider the visual representation for these things. We all know how binomia
[width=0.5] /PhyAI/chapter6/6_9*.png* Relationship between $N_\uparrow, S, U, M$ If our two-state paramagnet starts moving from sta
[width=0.8] /PhyAI/chapter6/6_10*.png* The graph of (U, S) What can we see here? Recall the definition of temperature (De
[width=0.8] /PhyAI/chapter6/6_11*.png* (U,T) and (T,M) graphs By sketching our data on graphs, the fact that $M \propto \dfrac{1}{T}$ st
$\approx k_B[N \ln N - N_\uparrow \ln N_\uparrow - (N - N_\uparrow) \ln (N - N_\uparrow)]$ The temperature function is: $1 \over T = \left(\frac{\partial S}{\partial U}\right)_{N,V,B} = \frac{\partial S}{\partial N_\uparrow} \cdot \frac{\partial N_\uparrow}{\partial U} = \frac{\partial S}{\partial N_\uparrow} \cdot \left(\frac{\partial U}{\partial N_\uparrow}\right)^{-1} = k_B[$
$N_\downarrow = \frac{N+U/\mu B}{2}$
We obtain:

I rearrange some terms here: $\exp\left(\frac{2\mu B}{k_B T}\right) = \frac{N-U/\mu B}{N+U/\mu B}$
$\Leftrightarrow \frac{N\mu B - U}{N\mu B + U} = \exp\left(\frac{2\mu B}{k_B T}\right)$
$\Leftrightarrow N\mu B - U = (N\mu B + U)\exp\left(\frac{2\mu B}{k_B T}\right)$
$\Leftrightarrow N\mu B\left[1 - \exp\left(\frac{2\mu B}{k_B T}\right)\right] = U\left[1 + \exp\left(\frac{2\mu B}{k_B T}\right)\right]$
$\Leftrightarrow U = N\mu B\left[\frac{1-\exp(2\mu B/k_B T)}{1+\exp(2\mu B/k_B T)}\right]$ From the definition of $\tanh x$ function: